PMAF Annotator — Instructions

1 Start the program

macOS: **right-click** START_HERE.command → **Open With→ Terminal**.

Windows: double-click START_HERE.exe. If SmartScreen objects: *More info → Run anyway*.

- Run only one copy at a time; a second launch refuses to start.
- To shut off the program, exit from the terminal, not just the browser.

2 Create a project from a video

Drop or upload an AVI/MP4 onto the upload box. Preparation extracts every frame and reads the burned-in UT clock. When it's *ready*, click **Annotate** to start the labeling.

3 The annotation loop

1. **(N)** Create a track. **One track = one physical PMAF**. You have to create a track before labeling each PMAF.
2. Drag the **circle tool** around the PMAF to label.
3. **(S) Snap**: fits a pixel mask to the bright aurora inside your circle. Move the threshold slider to update how much light the mask covers; fix details with the brush.
4. **Opacity**: changes the opacity of the pixel mask so you can see the underlying PMAF clearly.
5. (**A) Accept & Advance**: stores the frame, moves to the next frame, and copies the circle forward (you may still need to adjust the circle). You must click **Accept** on the frames you want to keep in order for them to be exported.
6. (**V) Absent**: the form is not visible on this frame.
7. (**Q) Bad Frame**: whole frame unusable (clouds, glare, corruption); press again to undo.
8. **Sky views**: (**D)** toggles **Original / Enhanced**. Enhanced view changes the color space you are looking at so it highlights dimmer lights, might help with labeling in darker frames.
9. **Advance step**: determines how many frames you are skipping in between. You can choose to label every frame, every two frames, every five frames, or every ten frames.
10. **Track Lifecycle**: when a track is selected, press **Start here** to indicate that a PMAF has appeared, and press **Ends here** when that PMAF disappears. Press **Delete Track** if a track is added by mistake.
11. Use **->** and **<-** to navigate between frames. The frame count is in the bottom-right corner; you can also navigate to specific frames using **Go To…**
12. A frame exports only when every track is accepted or absent on it.

4 Saving and quality

- Everything **autosaves continuously** (indicator in the top bar), no export needed to keep your work. Relaunching the tool reopens your project on the exact frame, track, and view you left. All work is saved in the *project folder.*
- When you **Export,** everything is saved in the *project folder.* That is what we need for model training.